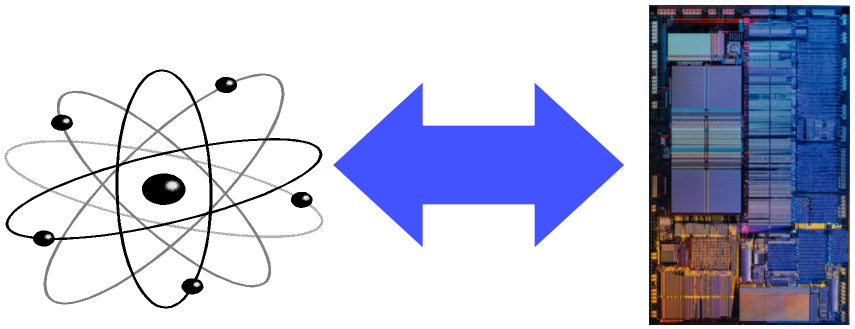
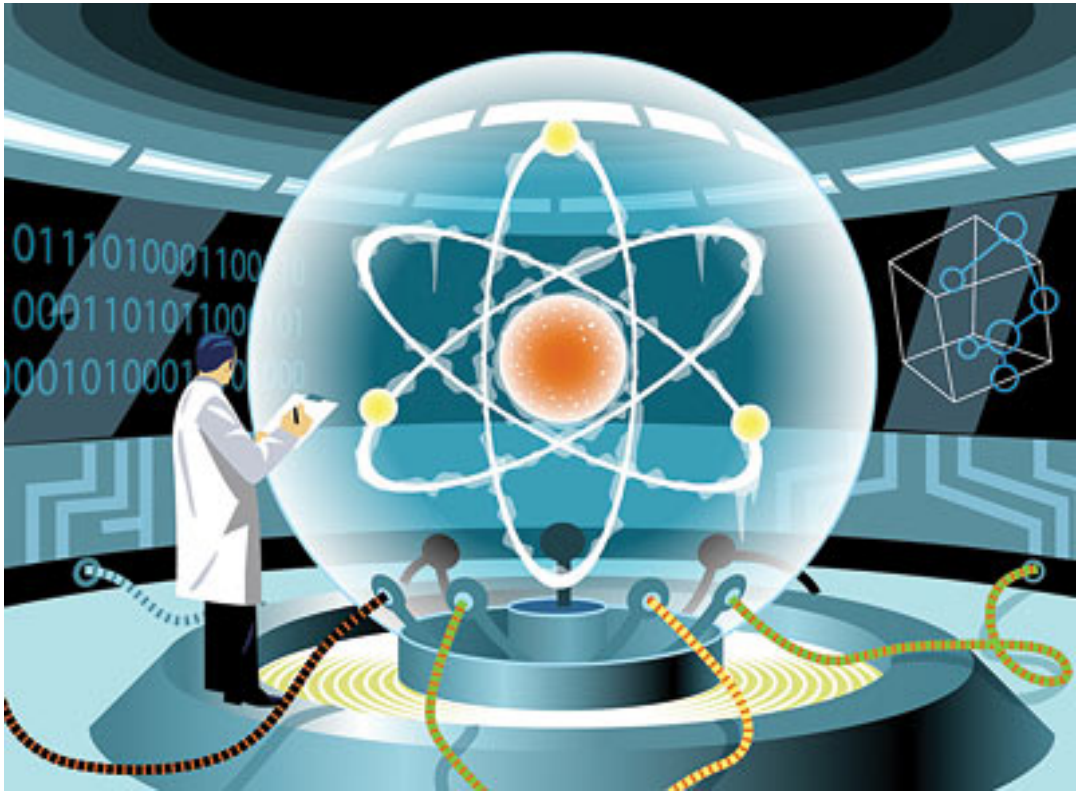
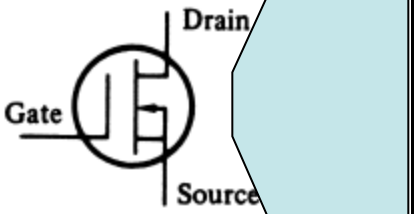
Quantum Computing Architectures

CSE 471 Lecture
May 31, 2007
Dave Bacon

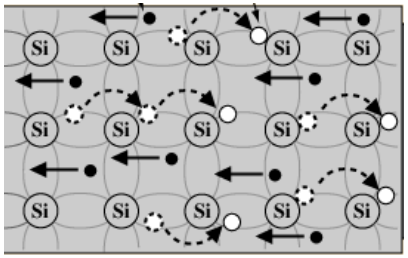


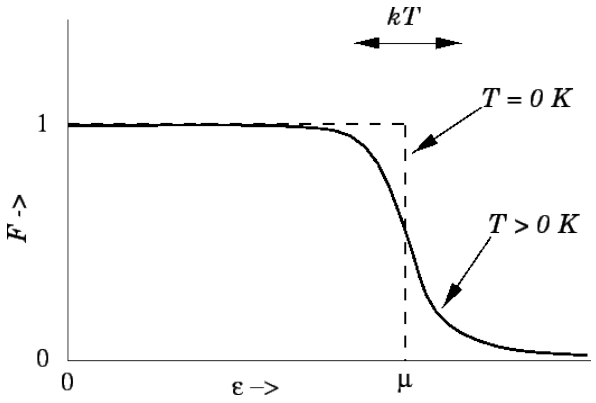


Behind the Bit

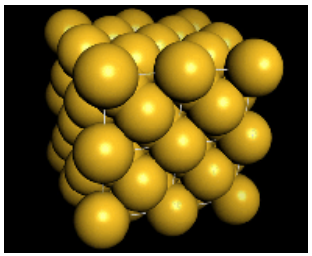


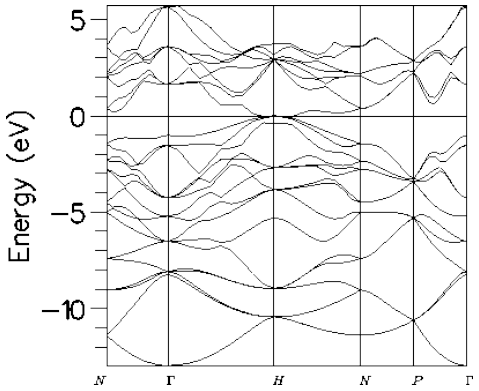
bit
s





noisy
bits





quantum bit

Bit, Pbit, and Qubit

	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent 1 *1-p* percent	0 $\lvert\alpha\rvert^2$ percent 1 $\lvert\beta\rvert^2$ percent
Evolution:	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$	$\begin{bmatrix} u & v \\ w & x \end{bmatrix}$
	deterministic	stochastic	unitary

Quantum Computers

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

×

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

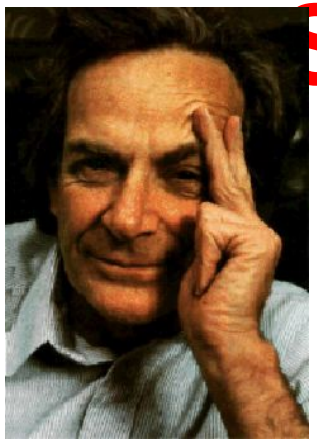


Peter Shor

1994

- Quantum computers can efficiently factor whole numbers (and compute discrete log.)
- No known classical efficient algorithm.
- Breaks RSA, Diffie-Hellman cryptosystems.

Quantum Simulations

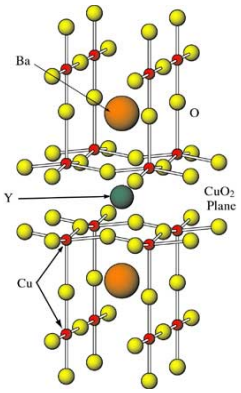


Richard Feynman

Simulating quantum systems on classical computers is hard

Simulating quantum systems on quantum computers is easy

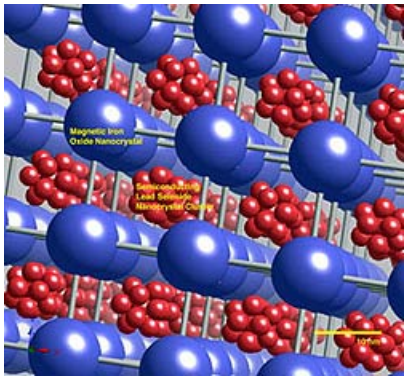
Quantum algorithms for simulating quantum processes



high temperature superconductors



biochemistry



novel materials

Architectures For Quantum Computers

Small scale (6-8 qubit) quantum computers have been built.

Systems-level challenges to achieving scalable, fault-tolerant quantum computation



1st quantum computing architecture “book”

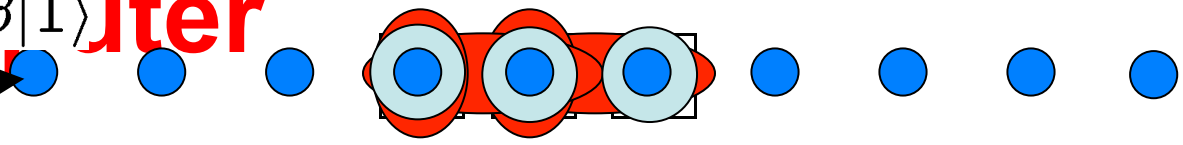
“Quantum Computing for Computer Architects”, Tzvetan Metodi and Frederic Chong (2006)

Available (free from UW!) at
http://www.morganclaypool.com/doi/abs/10.2200/S00066ED1V01Y200610CAC001

The Ideal Quantum Computer

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

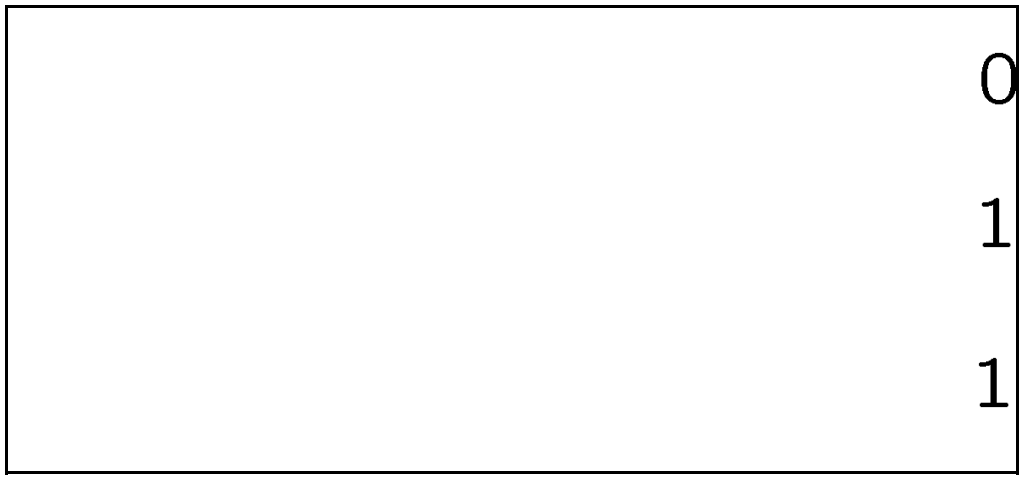
qubits



$$\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle - |111\rangle)$$

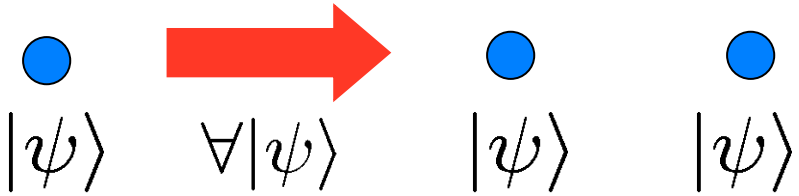
Quantum algorithm

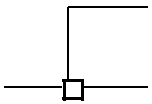
1. Initialize
2. Series of unitary gates
3. Measurement



“Unique” Quantum Challenges

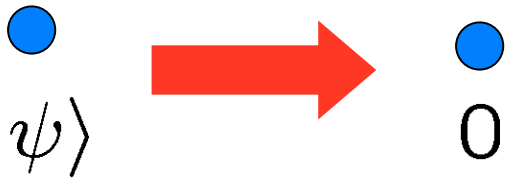
Quantum info cannot be copied:





“no fan-out”

Quantum info likes to become Classical:



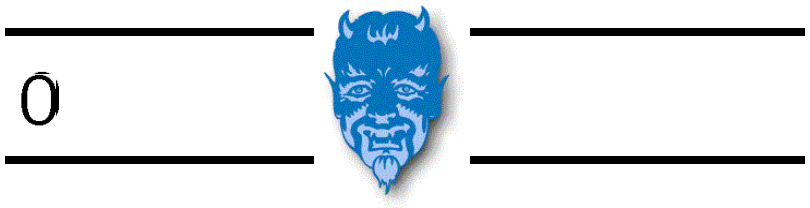
010001110101..



01001111000 1..

“noisy channel”

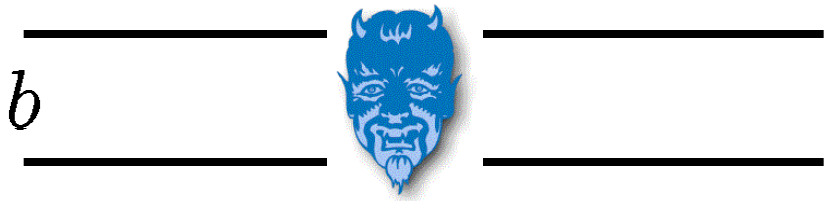
Classical Solutions



Probability of bit flip = *p*

Solution: redundancy

$0 \rightarrow 000$
$1 \rightarrow 111$



$$\left.\begin{matrix}000\\001\\010\\100\end{matrix}\right\} \rightarrow 0$$

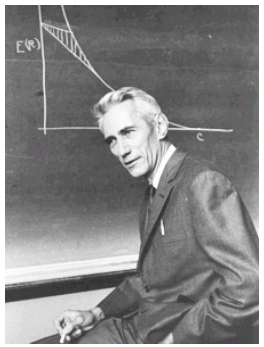
$$\left.\begin{matrix}111\\110\\101\\011\end{matrix}\right\} \rightarrow 1$$

Probability of failure =*3(1-p)p2+p3*

=3p2-2p3

<p if *p<1/2*

Use redundancy to protect information against noise



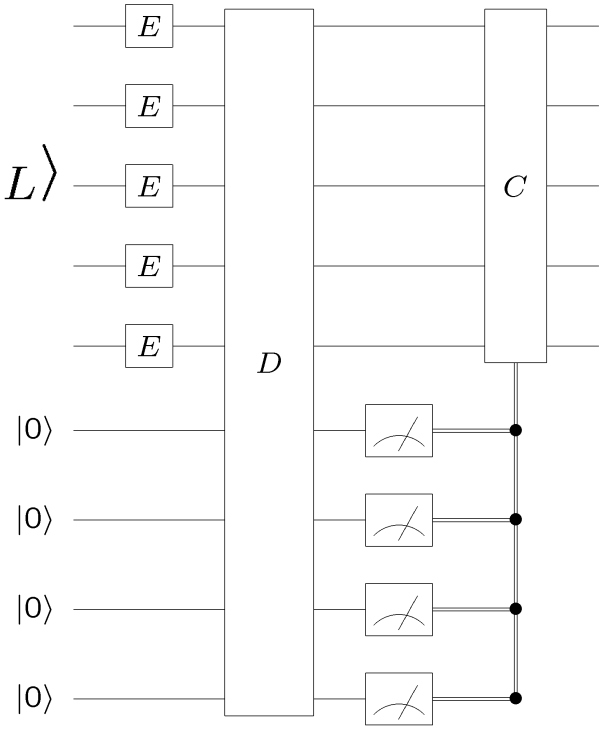
Claude Shannon 1948

“A mathematical theory of communication”

Quantum Error Correcting Codes

By encoding quantum information we can protect it from the effects of noise:

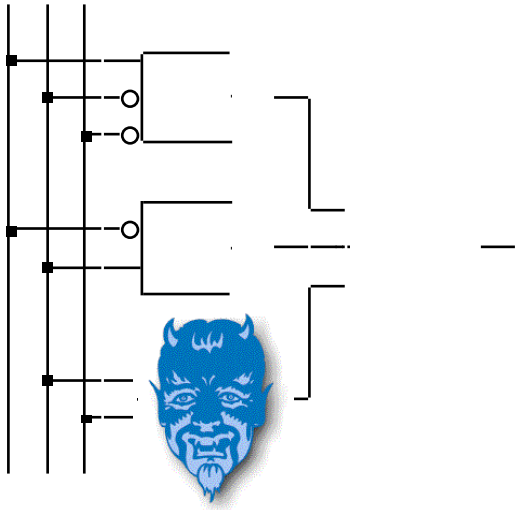
$\alpha|0_L\rangle + \beta|1_L\rangle$



$\approx \alpha|0_L\rangle + \beta|1_L\rangle$

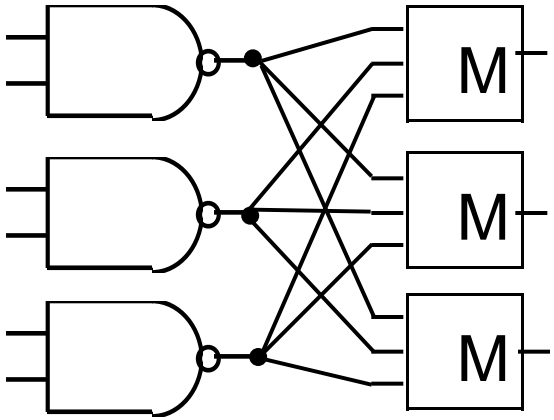
"bare" error rate = *p*
error rate on encoded = *O(p2)*

More Classical Problems



Classical gates fail with probability p

Solution: error correcting codes + fault tolerant construction

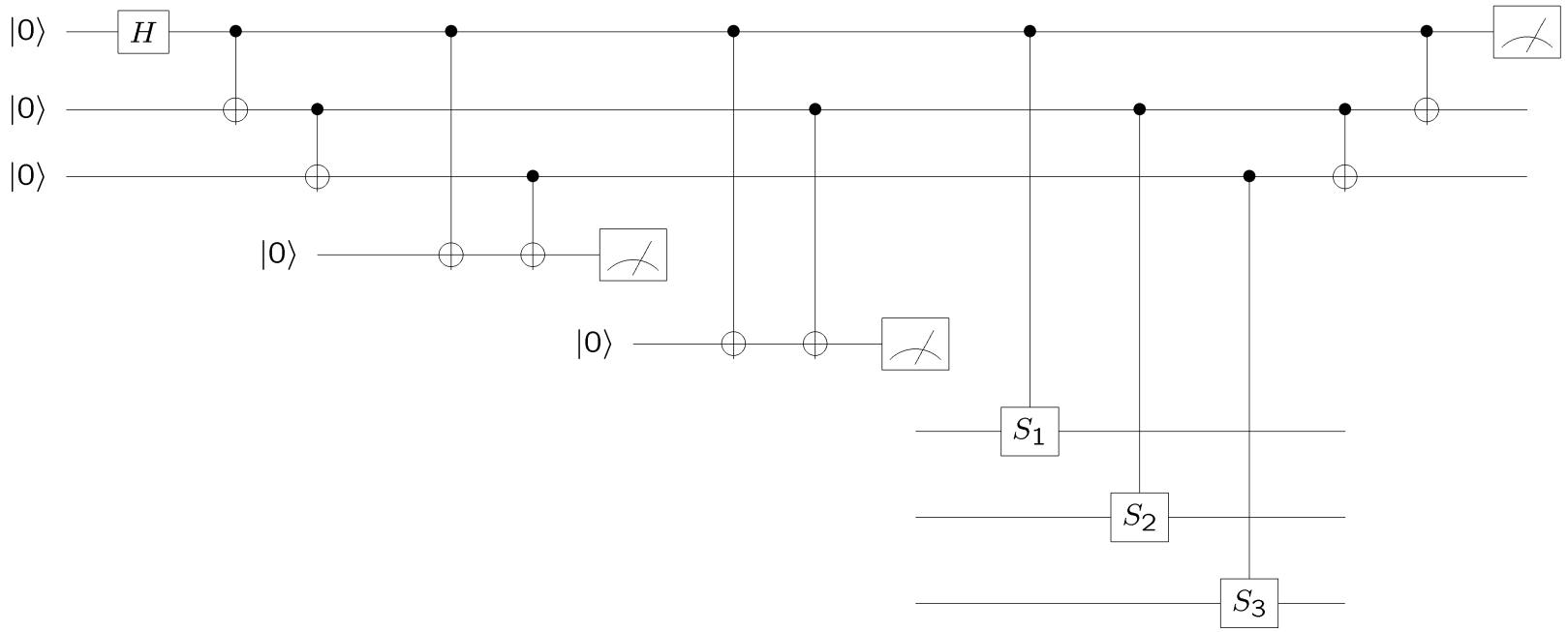




John von Neumann, 1956

“Probabilistic logics and the synthesis of reliable organisms from unreliable components”

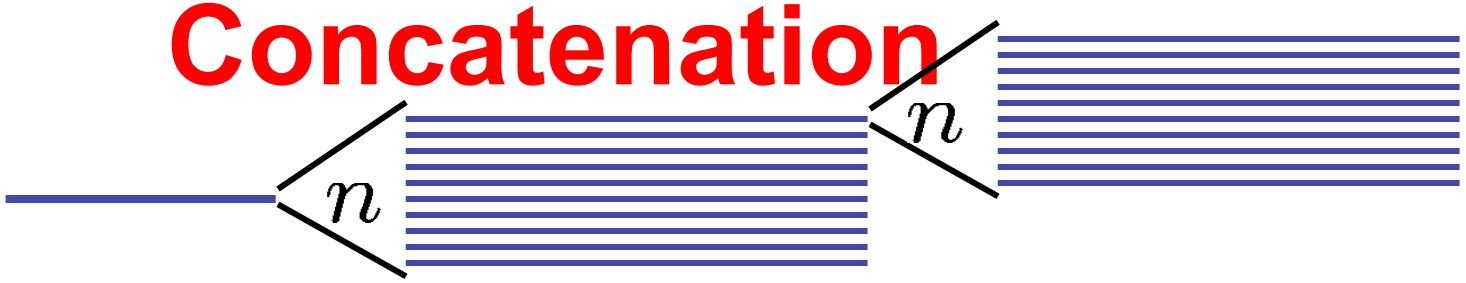
Fault-Tolerant Quantum Computation



Must design quantum circuits which succeed even when every element of the circuit (preparation, unitary gates, measurements) can fail.

Quantum error correction + Fault-tolerant procedures

Perfection Through Concatenation



failure probability

p $\quad$ $\frac{cp}{2}$ $\quad$ $\frac{c(cp2)}{2}$

At kth level of concatenation, probability of an error which destroys the encoded quantum information is

$$p_{fail} = \frac{(cp)^{2^k}}{c} \qquad \frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k} \qquad p_{th} = \frac{1}{c}$$

and uses $N_{qubits} = n^k$ qubits

$$N_{qubits} = \left[\log_{\frac{p}{p_{th}}} \left(\frac{p_{th}}{p_{fail}}\right)\right]^{\log_2 n} = O\left(\text{poly}\log \frac{1}{p_{fail}}\right)$$

Architectures For Quantum Computers

Systems-level challenges to achieving scalable, fault-tolerant quantum computation



Error Correction

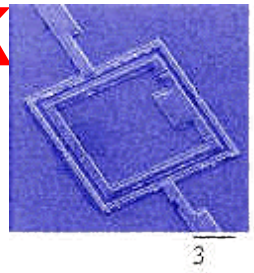
Designing computer architectures when components fail

The error correcting architectures must be massively (but fairly embarrassingly) parallel

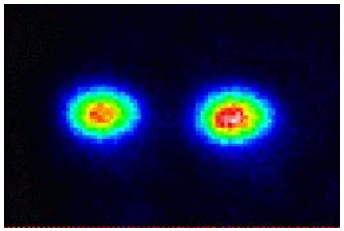
Building Block

qubits

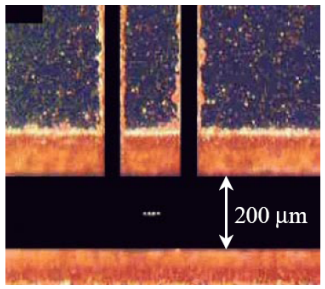
stationary
fixed location



Example:
superconducting qubits



walking
"slow"



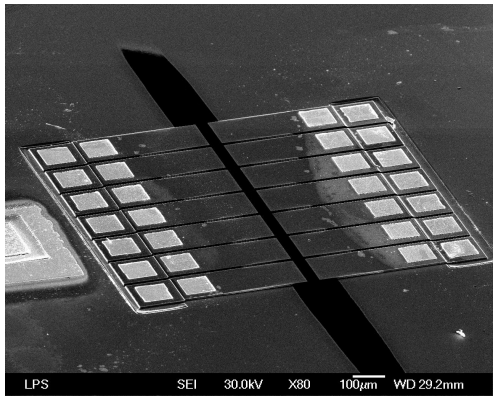
Example:
ion trap qubits

flying
"fast"

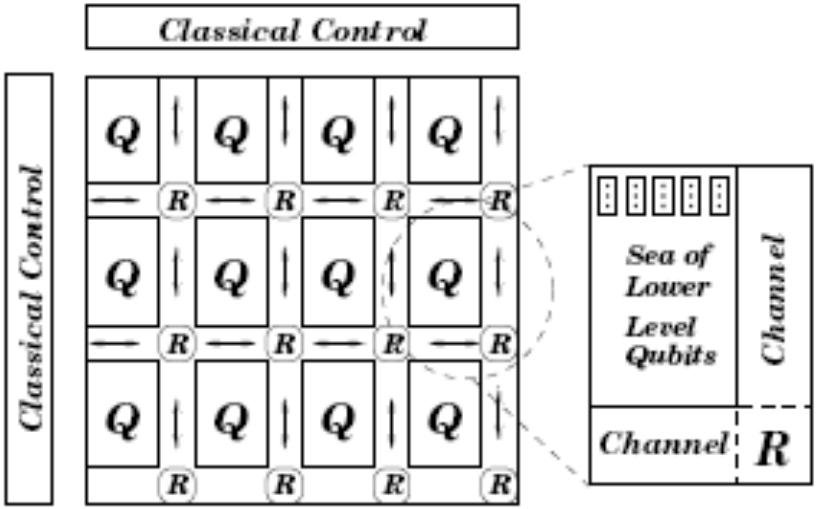


Example:
photons

Sketch Ion Trap QC

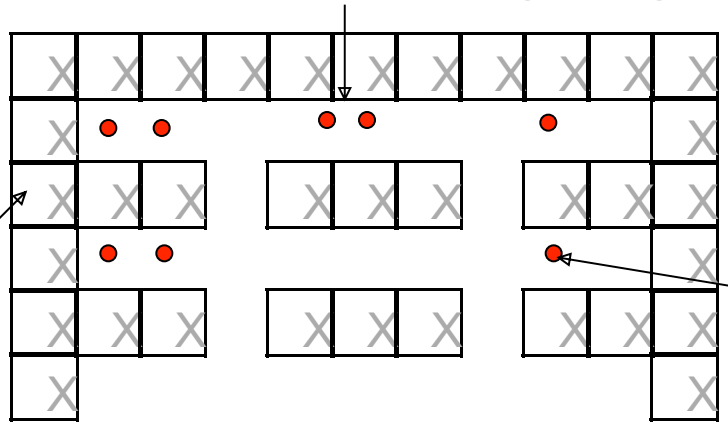


http://iontrap.physics.lsa.umich.edu/research/trap



(from Metodi et al, MICRO-38)

Interaction "trapping" regions

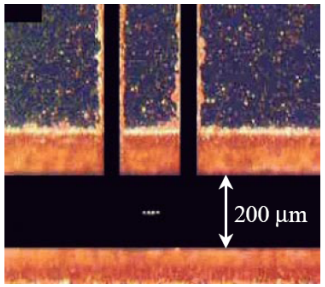


"electrodes"

ions

Time Scales

walkin g “slow”



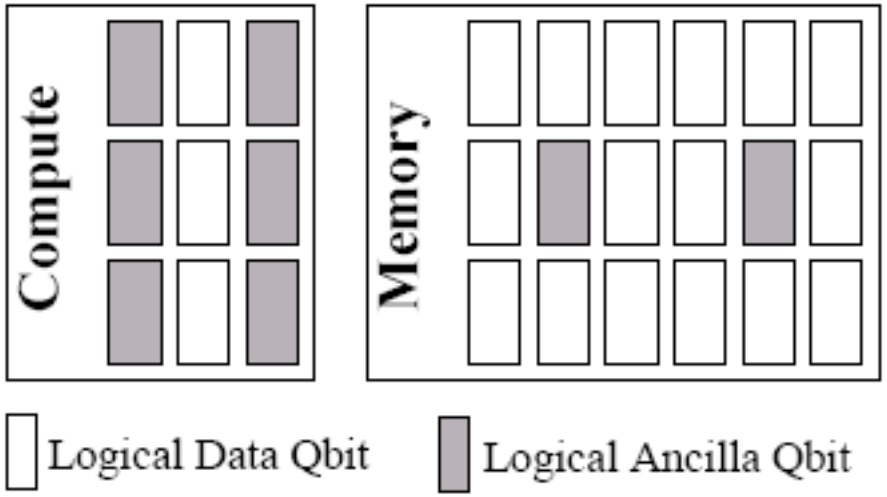
Example:
ion trap qubits

Operation	Time (µs)	Failure rate (today)	Failure rate (future)
single qubit gate	1	10-4	10-8
two qubit gate	10	10-2	10-7
measurement	200	10-2	10-8
movement	20	10-3/ µm	10-7/ µm
cooling	200	?	?
wait	107 / error		

Quantum Memory Hierarchy

Memory error rates << Errors during computation

Error correction more frequent during computation

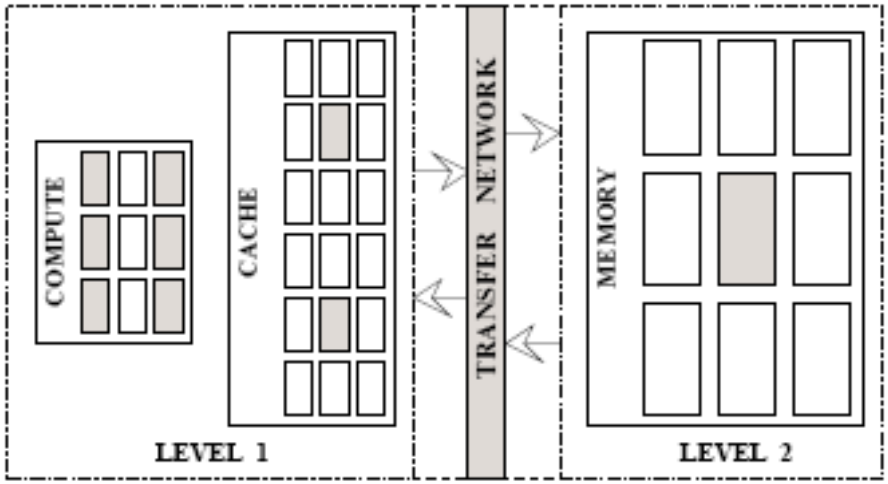


More ancillas (used for error correction) in compute region

Tradeoffs versus Size (Area)

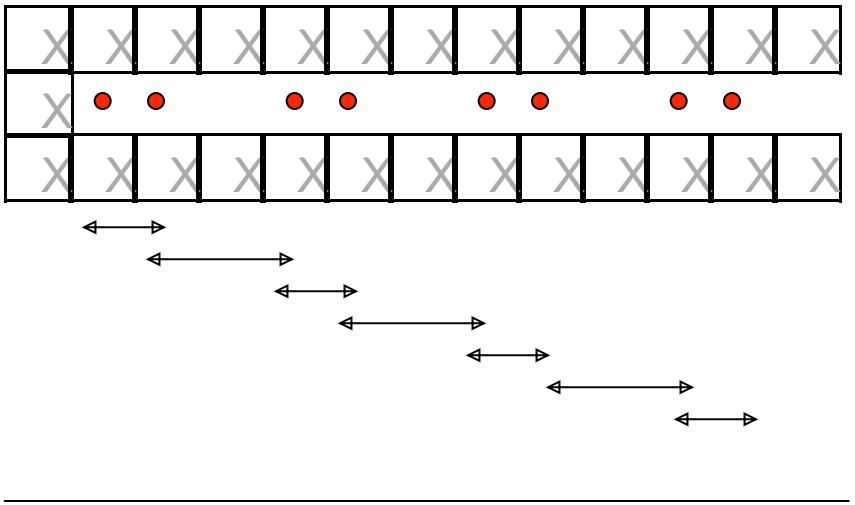
Quantum Memory Hierarchy

Slow memory / Fast compute leads to standard cache ideas…



except…

Communication



Distance k

Pf=
(p0)k

Exponential
decrease
in distance

Could perform error correction
every k0 steps….slow!

Quantum Teleportation

Alice wants to send $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ to Bob





$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

1 qubit

X

Y

2 bits





$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ ○ $|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ ○ $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

Alice performs operation on her state and half of entangled system and sends outcome

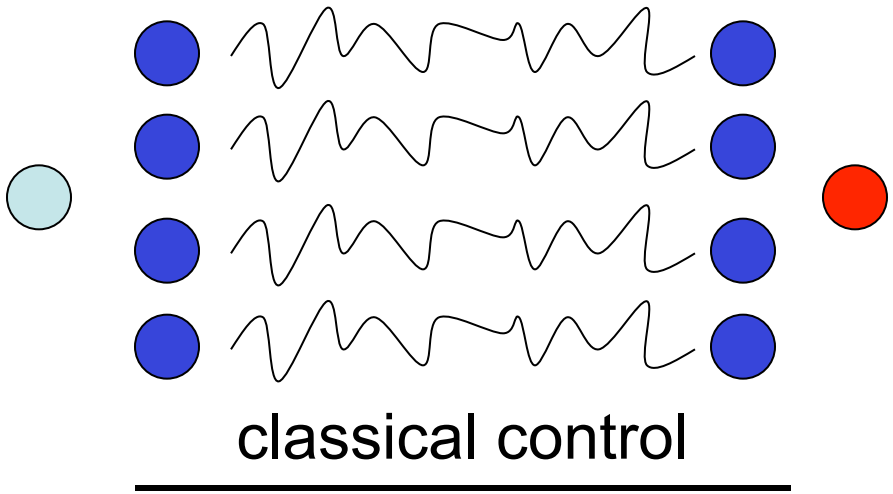
Conditional on received bits, Bob performs manipulation of his half of entangled system

Moving Qubits

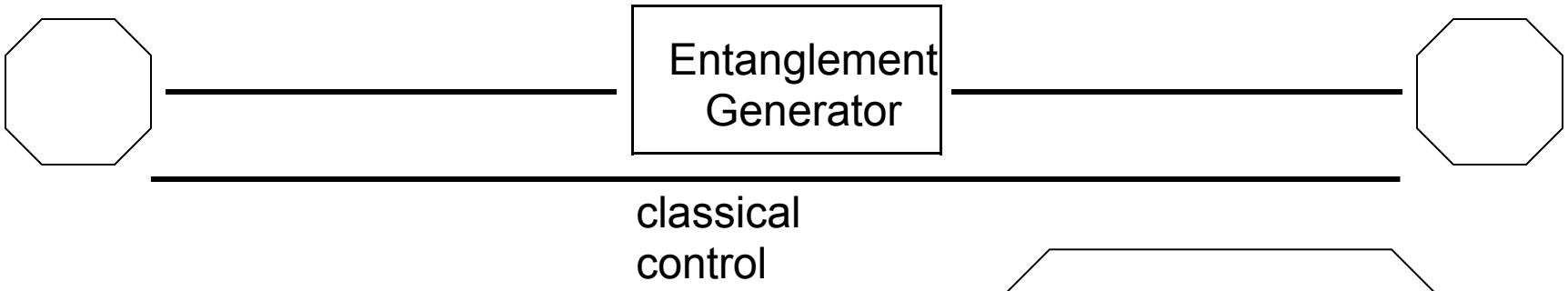
Teleportation channel



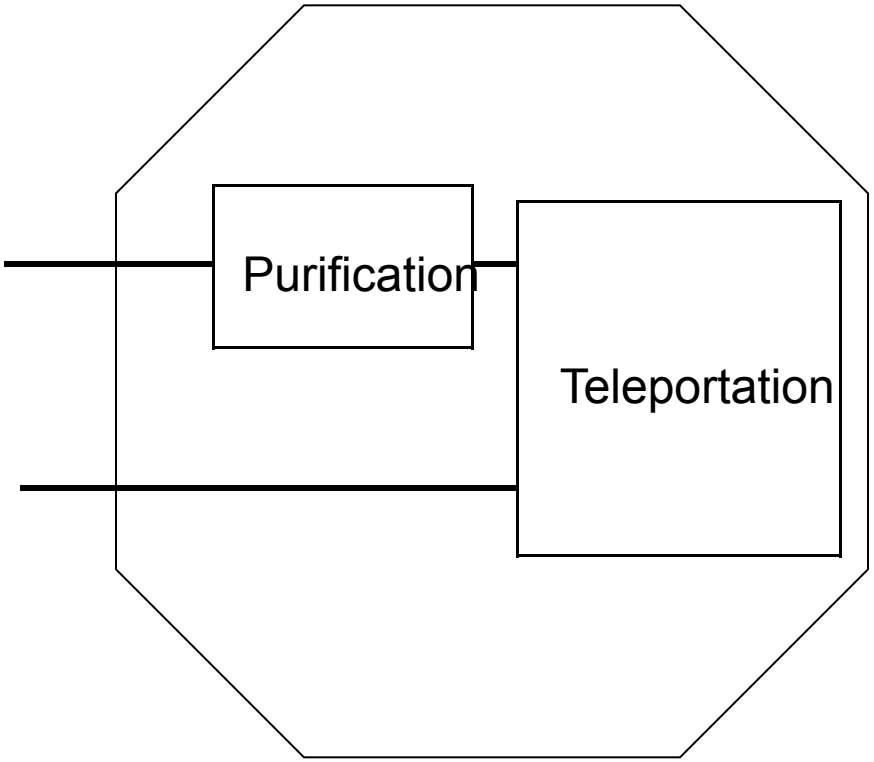
Entanglement distillation



Teleportation Units



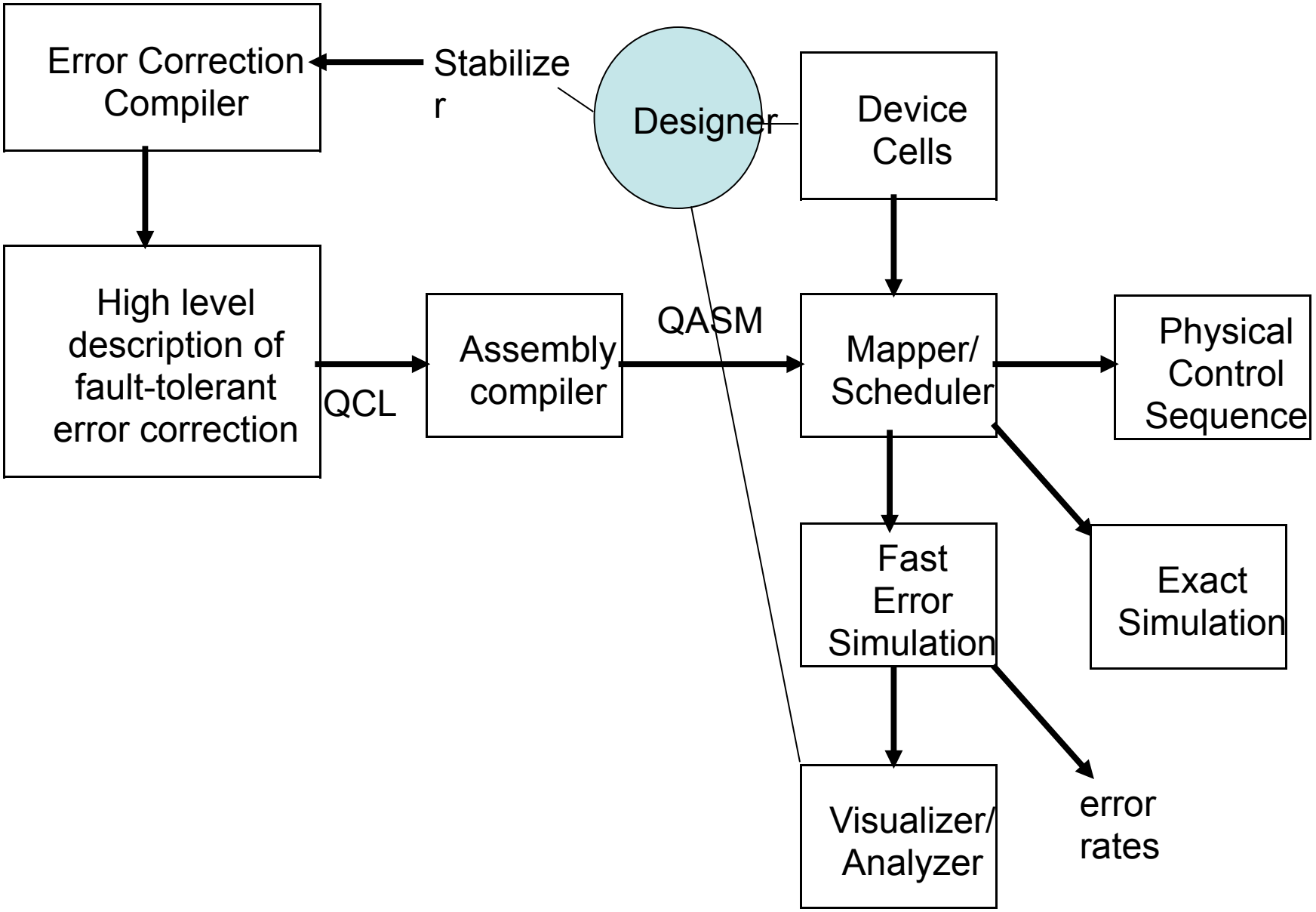
Parallel distribution of entanglement
+
Purification
=
On demand communication,
i.e. fast!



Quantum Architectures

New generation of tools in order to aid in the design of quantum computing architectures….

Tools!


Error Correction Compiler
Stabilizer
Designer
Device Cells
High level description of fault-tolerant error correction
QCL
Assembly compiler
QASM
Mapper/ Scheduler
Physical Control Sequence
Fast Error Simulation
Exact Simulation
Visualizer/ Analyzer
error rates

High Level Languages

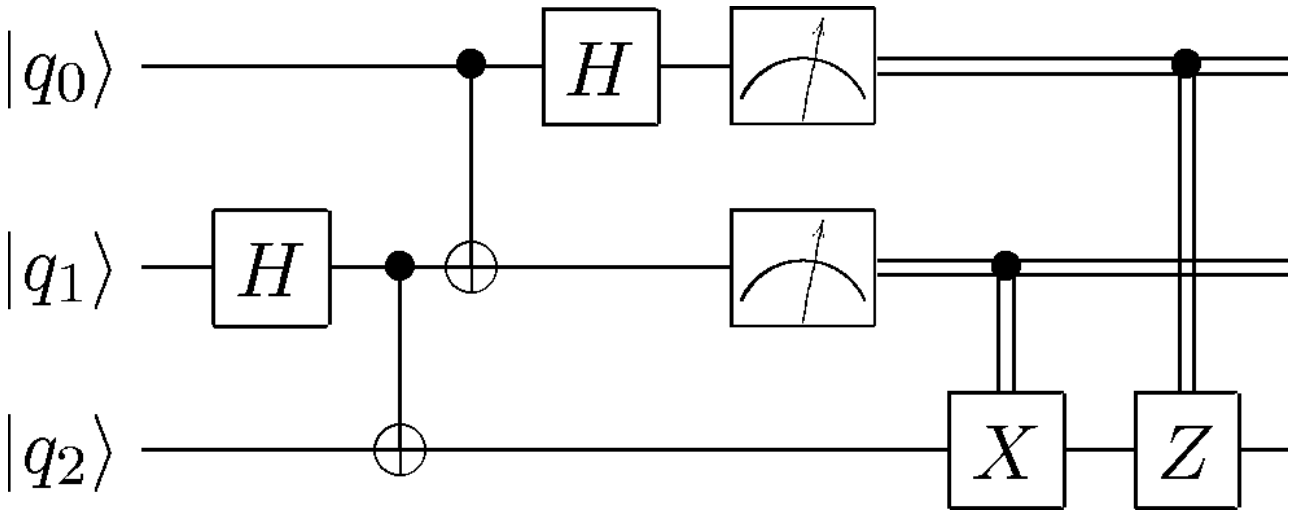
Example in QCL (http://tph.tuwien.ac.at/~oemer/qcl.html)

```
operator dft(qureg q) { // main operator
const n=#q;                  // set n to length of input
int i; int j;              // declare loop counters
for i=0 to n-1 {
   for j=0 to i-1 {          // apply conditional phase gates
      CPhase(2*pi/2^(i-j+1),q[n-i-1] & q[n-j-1]);
   }
   Mix(q[n-i-1]);            // qubit rotation
}
flip(q);                        // swap bit order of the output
}
```

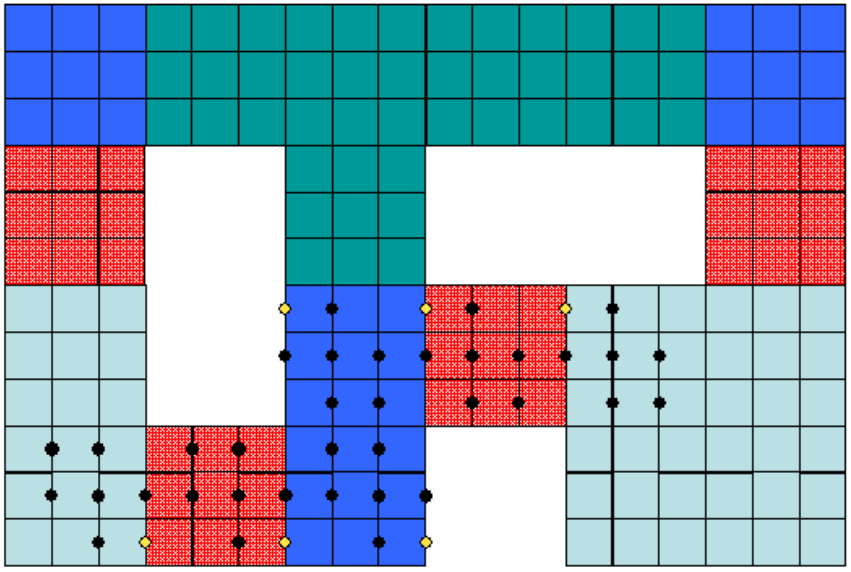
Quantum Assembly Language

```
qubit q0
qubit q1
qubit q2
h q1        # create EPR pair
cnot q1,q2
cnot q0,q1    # Bell basis measurement
h q0
nop q1
measure q0
measure q1
c-x q1,q2    # correction step
c-z q0,q2
```

QASM



Schedule
r



Z/X 1-EC

Example from Chuang group at http://www.media.mit.edu/quanta

Further Resources

Quantum Computing For Computer Scientists (David Mermin):
http://people.ccmr.cornell.edu/~mermin/qcomp/CS483.html

“Standard” Textbook:
Quantum Computation and Quantum Information
by Michael Nielsen and Isaac Chuang

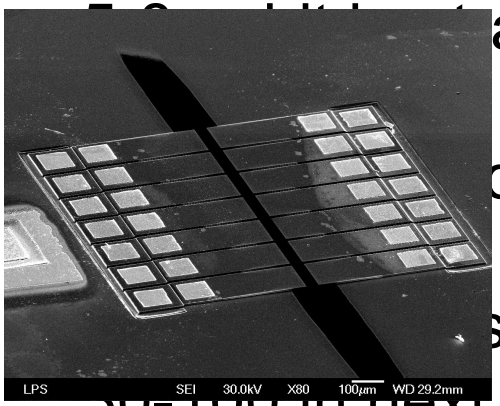
ISCA 2002 Tutorial Page (Oskin and Chong):
http://www.cs.washington.edu/homes/oskin/quantum-tutorial/

Quantum Architectures



Today: a taste

On the verge of becoming extremely important



ap machines

d ion traps

scale up to traps with

few year

Picture from